REPUBLIC OF SLOVENIA
**MINISTRY OF THE ENVIRONMENT AND SPATIAL PLANNING**
SLOVENIAN ENVIRONMENT AGENCY

PODNEBNI SKLAD

# Quality Assessment of Surface Current and Wave Measurements from the High-Frequency Radar System in the Gulf of Trieste

*Section 1:*

## Collecting and editing metadata from HFR GOT

Author: Marta Curri

Ljubljana, July 2026

# 1 Introduction

Due to wind and buoyancy forcings the circulation processes in the ocean surface tend to be the strongest and most unstable. High frequency (HF) instruments that operate within the radio bands of the electromagnetic spectrum (3 MHz - 50 MHz) use backscattered signals (*Bragg scattering*) from the ocean surface to map surface currents, wind, and waves.
Long-term deployments of HF radars (HFR) facilitate the characterisation of coastal circulation patterns, including wind-driven and tidal currents, as well as buoyant flows associated with river discharge. These observations also contribute to broader studies of marine ecology and coastal processes.

The Adriatic Sea is one of the hotspots of climate change, particularly the northern Adriatic shelf, bounded to the south by the Ancona–Zadar line. Due to its shallow depths, this area is exposed to large temperature fluctuations and strong currents, which are intermittently driven by the basin's dominant winds, the Bora and the Sirocco.
Because of these conditions, it is essential to monitor circulation in the northern Adriatic and in the Gulf of Trieste (GOT) over long, climate-relevant timescales.
To this end, a HFR system for measuring surface currents and waves was established in the GOT through cross-border cooperation between Slovenia and Italy.

This document evaluates metadata on the configuration of the High Frequency Radar - Northern Adriatic (HFR-NAdr) system for the period 2015–2026. It covers station locations, operating frequencies, temporal and spatial resolution, measured variables (currents, waves), data availability, and quality control (QC) algorithms.

Information was gathered from publicly available sources (websites, ERDDAP metadata, published articles) and supplemented with operator-confirmed values where available.
This data was collected, processed, and made freely available by The Regional Environmental Protection Agency of Friuli Venezia Giulia (ARPA FVG), The Environmental Agency of the Republic of Slovenia (ARSO), The National Institute of Biology (NIB), and The National Institute of Oceanography and Applied Geophysics (OGS), and the programs that contribute to it.

# 2 Network Overview

The NAdr network is a Wellen Radar (WERA) HF Radar System, which is an example of a beam-forming (BF) radar. The network comprises four antennas located across Italy and Slovenia: AURI, TRI1, IZOL, and PIRA.
Figure 2 illustrates the study area and the locations of the four stations, while Table 1 provides an overview of the system with information on the operating frequencies and the spatial and temporal resolutions.

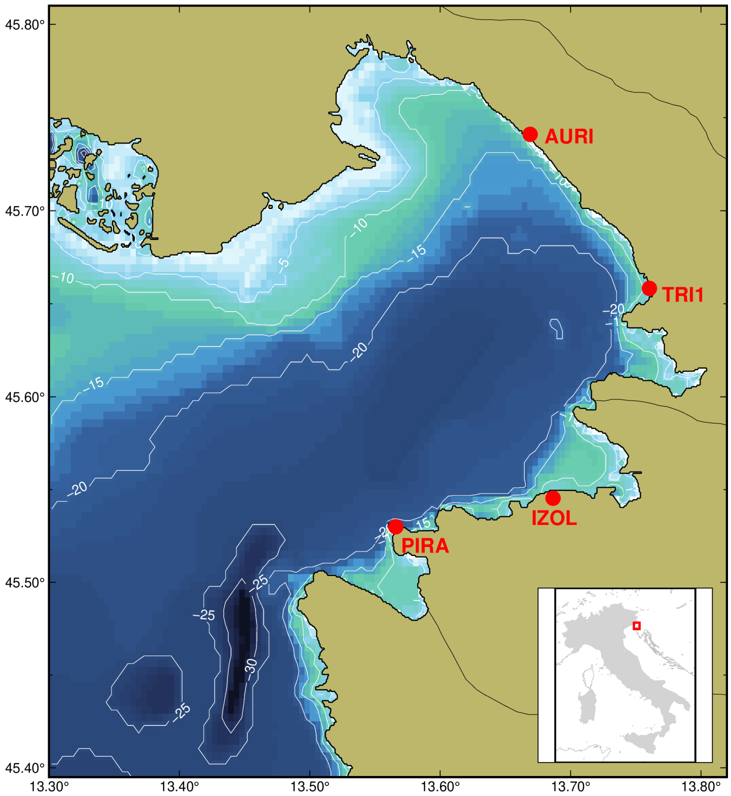


**Figure 2:** Bathymetry (GEBCO) of the Gulf of Trieste (the contour interval is 5 m). The location of the four HFR stations, AURI, TRI1, IZOL, and PIRA, are marked by red dots.

| | |
|---|---|
| System Name | HFR - NAdr |
| Area | Northern Adriatic Sea - Gulf of Trieste |
| Operators | ARPA, ARSO, OGS, NIB |
| Radar System | WERA HFR System |
| Operating Frequency | 24.5 - 24.635 MHz |
| Spatial resolution | 1.5 km grid<br>(0.0135 ° in latitude; 0.0193° in longitude) |
| Temporal Resolution | 30 minutes |
| Radial Coverage | 35 - 40 km |
| Depth represented | upper 0.3 - 2.5 m |
| Time Period | (2015) 2021 - present |
| DOI | `https://doi.org/10.57762/8RRE-0Z07` |

**Table 1:** System overview of the HFR-NAdr.

# 3 Funding Projects

- TOSCA (Tracking Oil Spills and Coastal Awareness): HF radar experiments in the Gulf of Trieste (2012 - 2014)
- HAZADR (Strengthening common reaction capacity to fight sea pollution): Installation of WERA stations AURI and PIRA (2012 - 2015)
- Firespill (Interreg IT-HR): Installation of TRI1 station in Trieste (2021 - 2023)
- SHAREMED: Installation of IZOL station in Izola (2022)

# 4 Stations

| Station | **AURI** | **TRI1** | **IZOL** | **PIRA** |
|---|---|---|---|---|
| Operator | OGS | ARPA FVG | NIB | ARSO |
| Location | Aurisina, Italy | Trieste, Italy | Izola, Slovenia | Piran, Slovenia |
| Coordinates | 13.669°E<br>45.741°N | 13.7605°E<br>45.6583°N | 13.6866°E<br>45.5454°N | 13.566°E<br>45.53°N |
| Frequency [MHz] | 24.635 | 24.525 | 24.5 | 24.525 |
| Operational Since | 2015 | 10/07/2023 | 01/03/2023 | 2015 |
| Status | Active | Active | Active | Active |

**Table 2:** Technical specifications of the four HFR-NAdr network stations.

## 4.1 Network configuration by period

| **Period** | **Active Stations** |
|---|---|
| 2014 | AURI, PIRA |
| 2015 | AURI only (PIRA down due to storm damage) |
| March 2023 - present | AURI, PIRA, IZOL |
| July 2023 - present | AURI, PIRA, TRI1, IZOL |

**Table 3:** Network configuration by period for the HFR-NAdr.

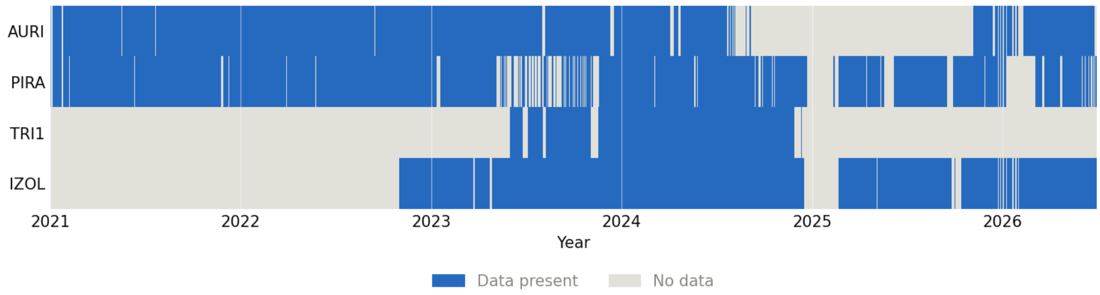


**Figure 3:** Measured hourly data availability per station, 2021–present, based on the radial current archives downloaded from the EU HFR Node. Blue indicates at least one valid measurement in that hour; gray indicates no data. Station start dates match Table 3. A shared gap of roughly ten months (late 2024 to late 2025) affects all stations and reflects an interruption in the archive itself rather than a per-station outage.

## 5 Data Availability

### Public data sources

| Source | URL | Content | Period |
|---|---|---|---|
| EU HFR Node ERDDAP | `erddap.hfrnode.eu` | Radials (4 stations) + Totals (v3) | 2021–present |
| EMODnet Physics ERDDAP | `erddap.emodnet-physics.eu` | Totals (`HFRADAR_NADR_Totals`) | 2021–present |
| OGS THREDDS | `dsecho.ogs.it/thredds/`[1] | Merged datasets | 2021–2024 |
| ARPA FVG | `meteo.fvg.it/radar_marino.php` | Currents + waves (visualization & archive) | – |

**Table 4:** Overview of Data Repositories, Available Products, and Temporal Coverage.

### 5.1 ERDDAP datasets (EU HFR Node)

| Dataset ID | Description |
|---|---|
| `EUHFR_NRTcurrent_HFR-NAdr-AURI_v3` | Radial velocities, AURI station |
| `EUHFR_NRTcurrent_HFR-NAdr-TRI1_v3` | Radial velocities, TRI1 station |
| `EUHFR_NRTcurrent_HFR-NAdr-IZOL_v3` | Radial velocities, IZOL station |
| `EUHFR_NRTcurrent_HFR-NAdr-PIRA_v3` | Radial velocities, PIRA station |
| `EUHFR_NRTcurrent_HFR-NAdr-Total_v3` | Total surface velocities |
| `EUHFR_NRTcurrent_HFR-NAdr-Total_v3_Last4Months_QC` | Total velocities, last 4 months only |

**Table 5:** Erddap datasets description.

**Note**: No wave datasets for HFR-NAdr are present on ERDDAP (search confirmed 2026-06-24).

# 6 Measured Variables

## 6.1 Surface currents

Each station measures radial current velocities toward/away from the antenna. Combining radials from two or more stations yields *total surface current vectors* on a regular spatial grid.

### 6.1.1 Radial Velocity

The dataset consists of maps of radial velocity of the sea water surface current collected in the GOT. Data are averaged over a time interval of 30 minutes. HFR measurements of ocean velocity are radial in direction relative to the radar locations and representative of the upper 0.3-2.5 meters of the ocean.
Table 6 provides an overview of the variables included in the dataset.

| Variable | Description | Unit |
|---|---|---|
| DRVA | Direction of radial vector away from instrument | degree |
| EACC | Radial accuracy of current velocity over coverage period | $\frac{m}{s}$ |
| EWCT | Surface eastward sea water velocity | $\frac{m}{s}$ |
| HCSS | Radial variance of current velocity over coverage period | $\frac{m^2}{s^2}$ |
| NSCT | Surface northward sea water velocity | $\frac{m}{s}$ |
| RDVA | Radial sea water velocity away from instrument | $\frac{m}{s}$ |
| | | **Threshold** |
| AVRB_QC | Average radial bearing quality flag | bearing range $= 0.0 - 360.0°$ |
| CSPD_QC | Velocity threshold quality flag | max. velocity $= 1.2\ \frac{m}{s}$ |
| MDFL_QC | Median filter quality flag | distance limit = 5.0 km; velocity median difference threshold $= 1.0\ \frac{m}{s}$ |
| OWTR_QC | Over-water quality flag | land/sea mask (GeoPandas naturalearth_lowres); not a numeric threshold |
| POSITION_QC | Position quality flag | no fixed numeric threshold (coordinate check/composite of the above) |
| QCflag | Overall quality flag | no fixed numeric threshold (coordinate check/composite of the above) |
| RDCT_QC | Radial count quality flag | minimum number of radial vectors = 25 |
| VART_QC | Variance threshold quality flag | maximum variance $= 1.0\ \frac{m^2}{s^2}$ |

**Table 6:** Grid Variables for Near Real Time Surface Ocean Radial Velocity by HFR-NAdr.

### 6.1.2 Total Velocity

The data set consists of maps of total velocity of the surface current in the GOT averaged over a time interval of 30 minutes. Surface ocean velocities estimated by HFR are representative of the upper 0.3-2.5 meters of the ocean.
Total velocities are derived using least square fit that maps radial velocities measured from individual sites onto a cartesian grid. The final product is a map of the horizontal components of the ocean currents on a regular grid in the area of overlap of two or more radar stations.
Table 7 provides an overview of the variables included in the dataset.

| **Variable** | **Description** | **Unit** |
|---|---|---|
| EWCT | Surface eastward sea water velocity | $\frac{m}{s}$ |
| GDOP | Geometrical dilution of precision | |
| NSCT | Surface northward sea water velocity | $\frac{m}{s}$ |
| UACC | Accuracy of surface eastward sea water velocity | $\frac{m}{s}$ |
| VACC | Accuracy of surface northward sea water velocity | $\frac{m}{s}$ |
| | | **Threshold** |
| CSPD_QC | Velocity threshold quality flag | maximum velocity = $1.0\frac{m}{s}$ |
| DDNS_QC | Data density threshold quality flag | minimum contributing radial velocities = 3 |
| GDOP_QC | GDOP threshold quality flag | maximum GDOP = 2.0 |
| POSITION_QC | Position quality flag | same as above (Table 6) |
| QCflag | Overall quality flag | same as above (Table 6) |
| VART_QC | Variance threshold quality flag | maximum variance = $1.0\frac{m^2}{s^2}$ |

**Table 7:** Grid Variables for Near Real Time Surface Ocean Total Velocity by HFR-NAdr-Total.

### 6.2 Waves

WERA radars measure wave parameters from the second order Doppler spectrum.

Wave data are not available on any public ERDDAP or THREDDS server. The EU HFR Node and EMODnet Physics publish currents only.

## 7 Key Publications

1. **Chiggiato et al. (2025)** — "Influence of wind stress and the Isonzo/Soča River outflow on surface currents in the Gulf of Trieste", *Ocean Science*, 21, 2197–2214. — Confirms 4 stations, wave measurements (Hs, direction), data period Oct–Nov 2023.

2. **Révelard et al. (2023)** — "Superstatistical analysis of sea surface currents in the Gulf of Trieste", *NPG*, 30, 515–531. — Analysis of data Jan 2021 – Oct 2022 (2 stations: AURI, PIRA). Documents transfer of PIRA management from NIB to ARSO in July 2022.

3. **Menna et al. (2022)** — "Coastal high-frequency radars in the Mediterranean – Part 2", *Ocean Science*, 18, 797–837. — Overview of HFR networks in the Mediterranean, including NAdr.

4. **Berta et al. (2021)** — "Multi-Platform, High-Resolution Study of a Complex Coastal System: The TOSCA Experiment in the Gulf of Trieste", *J. Mar. Sci. Eng.*, 9(5), 469. — Description of the 2012 TOSCA experiments.

# 8 Information Sources

| Source | Link |
|---|---|
| EU HFR Node — HFR-NAdr network | `https://www.hfrnode.eu/networks/hfr-nadr-2/` |
| ERDDAP — EU HFR Node | `https://erddap.hfrnode.eu/erddap/search/index.html?searchFor=NAdr` |
| EMODnet Physics ERDDAP — NAdr Totals | `https://erddap.emodnet-physics.eu/erddap/info/HFRADAR_NADR_Totals/index.html` |
| OGS THREDDS — NAdr-radar | `https://dsecho.ogs.it/thredds/catalog/radar/NAdr-radar/catalog.html` |
| ARPA FVG — Firespill news | `https://www.arpa.fvg.it/temi/progetti/progetti-europei/news/nuovo-radar-marino\-per-i-monitoraggi\-nel-golfo-di-trieste-grazie-\al-progetto-firespill/` |
| Chiggiato et al. (2025), Ocean Science | `https://os.copernicus.org/articles/21/2197/2025/` |
| Révelard et al. (2023), NPG | `https://npg.copernicus.org/articles/30/515/2023/` |
| NIB — HF radar (SHAREMED/IZOL installation) | `https://www.nib.si/mbp/en/oceanographic-data-and-measurements/other-oceanographic-data/hf-radar-2` |
| SHAREMED — official project site | `https://sharemed.interreg-med.eu/` |

**Table 8:** Relevant sources and data repositories for the HFR-NAdr network.